Wk 2 Additional reading

Foreword:

During Wk1 I worked on the code basics for the simulation whilst my lab partner Scarlett produced a robust literature review on the topic. This page is intended to summarise and notate the findings of that and similar readings. This review relied on [(n.d.a)](Binder 1976)(J M Sanchez 1978)[M. B. Yilmaz (2005)[Domb and Stoll (1977)](Delboo 2022)

Cluster theoretical overview (overview provided by Scarlett, rewritten here for convenience)

Scarlett wrote a quick summary of her theoretical findings which are renotated here.

She focused on characterising clusters in the Ising model, so outlined the following cluster observables:

- Cluster size distribution
- Cluster shape
- Spatial correlations
- Time autocorrelation
- Connection between clusters and correlations.

Cluster size distribution

$z(n) = \frac{N_{cluster}(n)}{N}$ Eq 2.1 Cluster size density \$N_{cluster}(n) = \$ Number of clusters of size n \$N = \$ number of spin states in ising model

$z(n)$ represents the "density" of clusters of a given size n in an ising model. Mean cluster size follows:

$< n >_{cluster} = \frac{\sum n^2 z(n)}{\sum n z(n)}$ Eq 2.2 Average cluster size

Which apparently has behaviour analagous to magnetic susceptibility The number of clusters per lattice point:

$< C >_{site} = \sum z(n)$ Eq 2.3 number of clusters per site

With the largest cluster fraction acting as an order parameter:

$f_{max} = \frac{n_{max}}{N}$ Eq 2.4

Note that at critical temperature, $z(n)$ approaches:

$z(n) \approx \frac{1}{n^{2+\frac{1}{\delta}}}$ Eq 2.5 Critical cluster density \$ = \$ critical exponent of magnetisation-field response (2D case has $\delta = 15$)

Cluster shape:

It is useful to define a "cyclomatic number" for any cluster (Domb and Stoll 1977):

$c = l - n + 1$ Eq 2.6 cyclomatic number $n = $ number of spins in a cluster $l = $ number of bonds between spins in a cluster

Given this, some idea of "compactness" may be defined and illustrated:

$A = \frac{c}{(n-1)^2}$ Eq 2.7 Coefficient of compactness

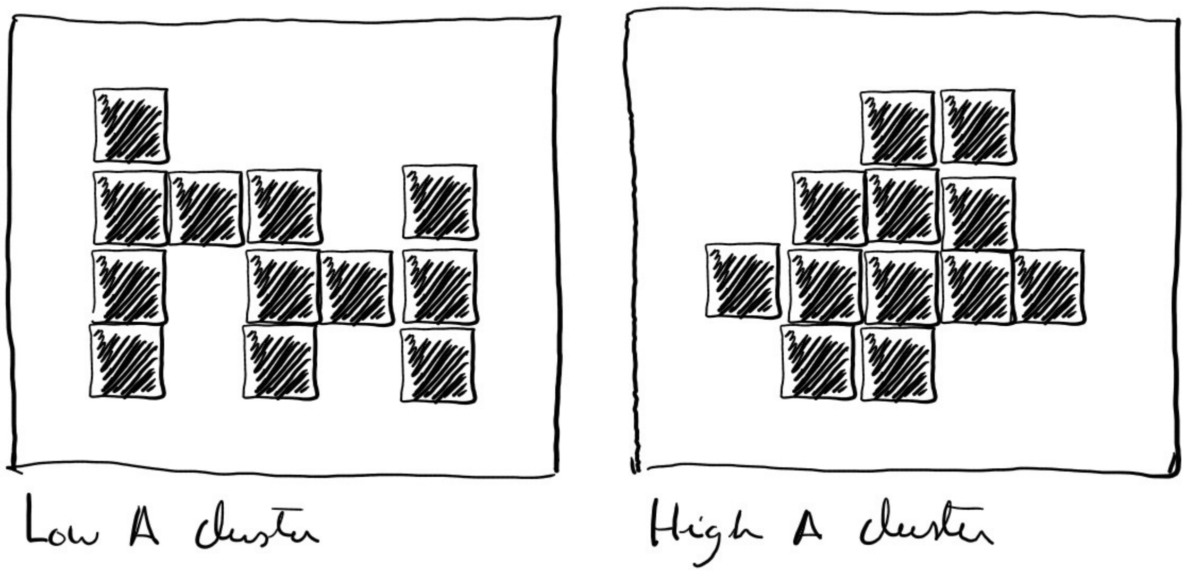

Figure 1: Fig 2.2 - Illustration of a non compact vs compact cluster

Spatial correlations:

Spin correlations are described by:

$C(r) =< s(0)s(r) >= \frac{1}{N} \sum s_i s_{i+r}$ Eq 2.8 equal time correlation function

At non critical temperatures this experiences rapid decay:

$c(r) \approx e^{\frac{-r}{\zeta}}$ Eq 2.9 Non critical correlation $ = $ correlation length

ζ is related to criticality, so as the model approaches criticality it grows, increasing the correlations between distant lattice points.

Time autocorrelation:

Time autocorrelation is analagous to spatial correlations as Eq 2.7, but across some time interval τ instead of space r:

$A(\tau) = \frac{<(M_t - \overline{M})(M_{t+\tau} - \overline{M})>}{(M_t - \overline{M})^2}$ Eq 2.10

and typically decays as:

$A(\tau) \approx e^{\frac{-tau}{\tau_c}}$ Eq 2.11

Connections between cluster and correlations:

Spin correlations and cluster connectivity are directly related as:

$< s_i s_j >= P(i <-> j)$ Eq 2.12 spin correlations and probability of connection between two points

Critical exponents papers:

Scarlett found 4 Critical exponent related papers which i need to go through and summarise.

CritExpLink1 (Burt 2024)

Initial impressions:

- computed crit exp α, β, γ for 3D IM
- trained deep learning alg on IM results to reduce necessary information of system

CritExpLink2 (Genzor 2023)

Initial impressions:

- Calculated critical exponents on "fractal lattice" w/$d = 1.792$
- used "higher order tensor renormalisation" to compute this

CritExpLink3 (Rieger and Young 1993)

Initial impressions:

- found crit exp for correlation length, specific heat, susceptibility, disconected susceptibility, magnetisation
- 3D model
- used finite size scaling to find
- seems to be most "general"

paper specifics:

Used a MC 3D ising model to generate a set of "sample models" and associated "replica models" over a range of temperatures. The replica models were used to decide when a model had equilibrated by comparing the magnetisations of 2 models; one which started all spin up, one which started all spin downs. When the magnetisations entered each other's uncertainties; equilibration was determined.

Each sample/replica measured:

- $< M >=$ average magnetisation
- $< M^2 >$
- $< E >=$ average energy
- $< E^2 >$

Which yield the following observables:

$C_{ave} = N\frac{(<E^2>-<E>^2)}{T^2}$ Eq 2.13 average specific heat capacity

$m_{ave} = | < M > |$ Eq 2.14 Average magnetisation

$\chi_{ave} = N\frac{(<M^2>-<M>^2)}{T}$ Eq 2.15 Average susceptibility

$\chi_{dis} = N < M >^2$ Eq 2.16

Which relate to their critical exponents as:

$t = T - T_c$ Eq 2.17 Temperature relative to critical temperature T_c

$T^2 C_{ave} = L^{\frac{\alpha}{v}}\overline{C}(tL^{\frac{1}{v}})$ Eq 2.18

$m_{ave} = L^{\frac{\beta}{v}}\overline{m}(tL^{\frac{1}{v}})$ Eq 2.19

$T\chi_{ave} = L^{2-\eta}\overline{\chi}(tL^{\frac{1}{v}})$ Eq 2.20

$\chi_{dis} = L^{\eta-\overline{eta}}\chi_{dis}(tL^{\frac{1}{v}})$ Eq 2.21

$\alpha =$ Specific heat critical exponent

$\beta =$ order parameter exponent

$v =$ Correlation length exponent

$\eta, \overline{\eta}$ describe "power law decay"

$\gamma = (2-\eta)v =$ susceptibility exponent

$\overline{C}, \overline{m}, \overline{\chi_{ave}}, \overline{\chi_{dis}}$ are scaling functions for their respective observables.

The paper found the critical exponents for C_{ave} by acknowledging that $\overline{C}$ has a maxima $x_0 = (T_0 - T_c)L^{\frac{2}{v}}$ for some T_0 where $T^2 C_{ave}$ is maximised; so by finding $T_0(L)$ over a range of L, the exponent v and critical temperture can be fitted with:

$T_0(L) = x_0 L^{\frac{-1}{v}} + T_c$ Eq 2.21

So the free parameters x_0, v and T_c can be found by polynomial fit.

The paper implies similar procedure may be used for the other observable parameters.

The paper then discusses these exponents in a wider context that i'm not fully versed in, and at the moment not fully interested in. The outline in measuring critical exponents was the main point of interest for this lab at this point.

CritExpLink4 (n.d.b)

Initial impressions:

- explanation of mean field theory approx

n.d.a. https://www.ggi.infn.it/talkfiles/slides/slides6163.pdf.

———. n.d.b. https://www2.ph.ed.ac.uk/~mevans/sp/sp10.pdf.

Binder, K. 1976. "'Clusters' in the Ising Model, Metastable States and Essential Singularity." *Annals of Physics*. https://doi.org/https://doi.org/10.1016/0003-4916(76)90159-7.

Burt, T. 2024. "Computing Critical Exponents in 3D Ising Model via Pattern Recognition/Deep Learning Approach." https://arxiv.org/pdf/2411.02604v1.

Delboo, M. 2022. "Using the Random-Cluster Model for Ising Model Simulations." https://repository.tudelft.nl/file/File_62c5834c-11be-4455-a96e-e739bc6d9efa?preview=1.

Domb, C, and E Stoll. 1977. "Shape and Size of Clusters in the Ising Model." *Journal of Physics A: Mathematical and General*. https://doi.org/10.1088/0305-4470/10/7/012.

Genzor, Jozef. 2023. "Calculation of Critical Exponents on Fractal Lattice Ising Model by Higher-Order Tensor Renormalization Group Method." *Physical Review E*. https://doi.org/10.1103/PhysRevE.107.034131.

J M Sanchez, D de Fointaine. 1978. "The Fcc Ising Model in the Cluster Variations Approximation." *Physical Review B*. https://journals.aps.org/prb/pdf/10.1103/PhysRevB.17.2926.

M. B. Yilmaz, Frank M. Zimmermann. 2005. "Exact Cluster Size Distribution in the One-Dimensional Ising Model." *Physical Review E*. https://doi.org/10.1103/PhysRevE.71.026127 .

Rieger, H, and A P Young. 1993. "Critical Exponents of the Three-Dimensional Random Field Ising Model." *Journal of Physics A: Mathematical and General*. https://doi.org/10.1088/0305-4470/26/20/014.